Agenda

- This Week
  - Class 14 – October 28th (2:00 – 3:00, 3:30 – 4:30)
  - Class 15 – October 29th (10:30 – 12:30)
  - Class 16 – October 30th (1:00 – 3:00)

- Starter Quiz

- Integration Architecture

- Thought Paper Presentation
  - Consumer Persona's – Students, Parents (Team 5)

- Integration Architecture (Cont'd)

Note:

- Mobile Phones – Do not bring to class or Completely Switch off
- If students have any specific topic they want to discuss, share the topic and we can try to schedule
- Please be seated by your teams

Starter Quiz

- Jiro turned 100 years yesterday – Who is "Jiro Ono" ?

Jiro Ono - Renowned Japanese Sushi Master

***Jiro Dreams of Sushi* – Watch the Award Winning Documentary**

- Regarded as the greatest living sushi master and chef
- Celebrated for his "**Relentless pursuit of perfection"**
- Owns "Sukiyabashi Jiro" a 10-seat, sushi-only restaurant located in a Tokyo Ginza subway station
- Dedication to craft - For the last close to 80 years spent his life refining the art of sushi-making, with a focus on meticulous preparation and achieving the perfect balance of rice, temperature, and fish.

Starter Quiz

- **What is the difference between "Tokenized Deposit" and "CBDC"**

Digital Assets - Stablecoins Vs Tokenized Deposits Vs CBDC's

	Stablecoins	Tokenized Deposits	CBDCs
Issuer	Licensed Institutions	Commercial Banks	The Central Bank
Trust Model	Inherited from backing assets	Issuer's balance sheet, backed by regulation	Legal mandate, ultimately backed by the tax raising powers of the government
Backing Model	100% backing, in a mix of • deposits held at banks • high-quality liquid assets • low-risk, longer-dated assets	Fractional reserve banking model, with capital and liquidity requirements set and monitored by regulator	The Central Bank Reserves
Claim	Claim on the issuer, or on the backing assets directly if held in trust.	Claim on the bank	Claim on the central bank and ultimately the government
Redemption Model	Often decentralized mint/burn capability	In branch, via online/mobile banking, or via ATM network	None (previously to gold)
Convertibility	Freely convertible at centralized or decentralized digital asset exchanges	Easily convertible to other deposits or card balances	Convertible at commercial banks or regulated stablecoin issuers

Core Banking Platform Engineering – Integration Architecture

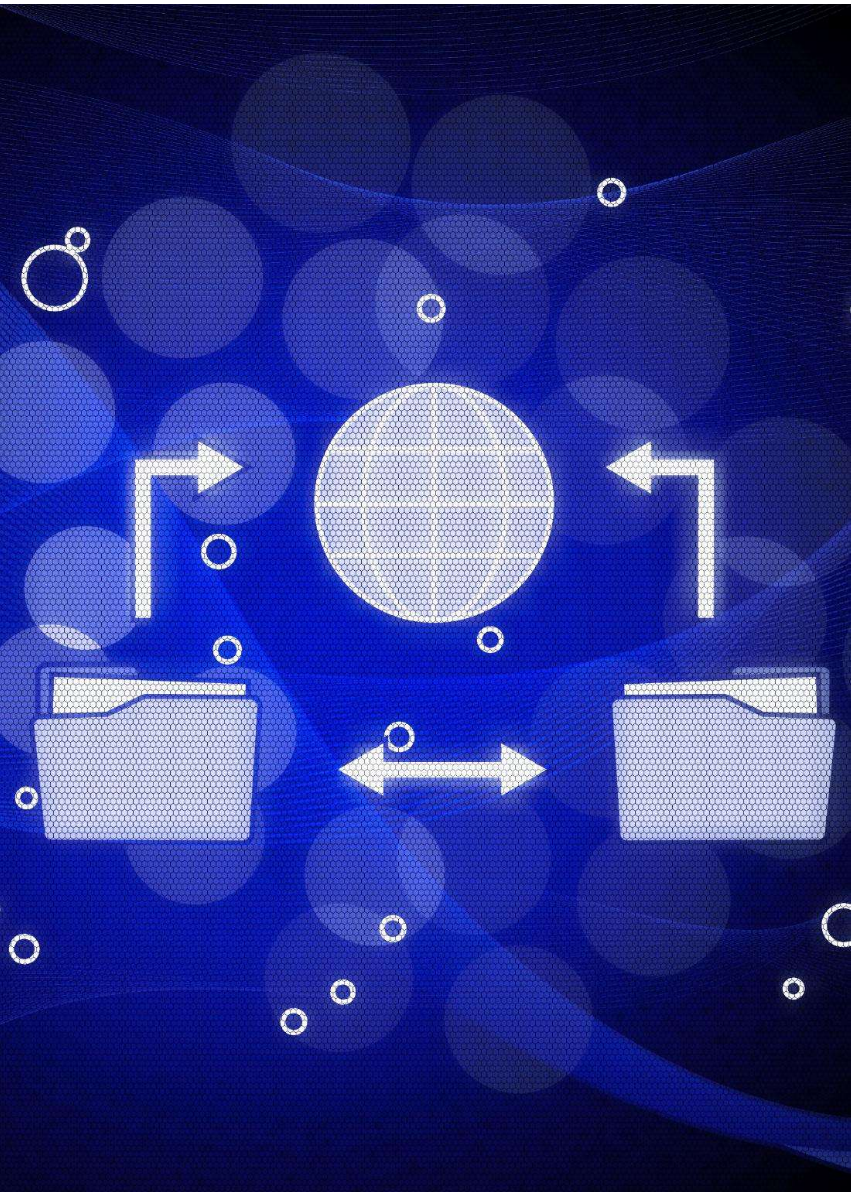
Integration Architecture – Topics Covered

- **Integration Patterns**
  - **Real Time Integration – API’s**
  - **Near Real Time Integration – Event**
  - **Batch/Bulk Integration – Files**

  Note: Agentic AI Integration will be covered in the AI Session Planned for Nov 11th

- **Industry Standard Interfaces**
  - **ISO 8583**
  - **ISO 20022**
  - **BIAN**
  - **Open Banking**

Real Time Integration - API



Real Time Integration - API

Instant Data Exchange

Real-time API integration enables immediate data sharing between systems, boosting responsiveness and reducing delays across platforms.

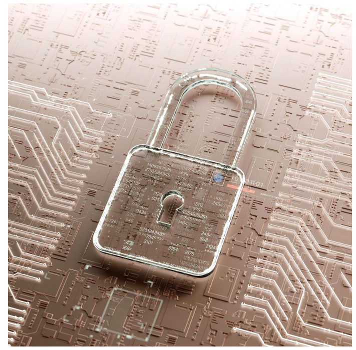
Dynamic Information Updates

APIs support dynamic updates, ensuring users always see the most current information whenever they interact with the system.

Improved Workflow Efficiency

Real-time integration streamlines workflows drives efficiencies and improves Consumer Experience

Real Time Integration - API Management and API Gateway



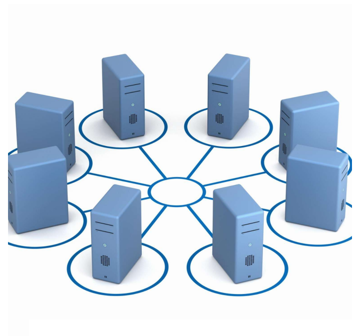
Secure API Creation and Publishing

API management enables organizations to securely create and publish APIs, ensuring controlled access.

The API Gateway serves as a central entry point for client requests, efficiently routing them to backend services.

They enable smooth integration, protocol translation, and centralized API management for modern applications.

Major API Gateways – WSO2, Apigee, Mulesoft



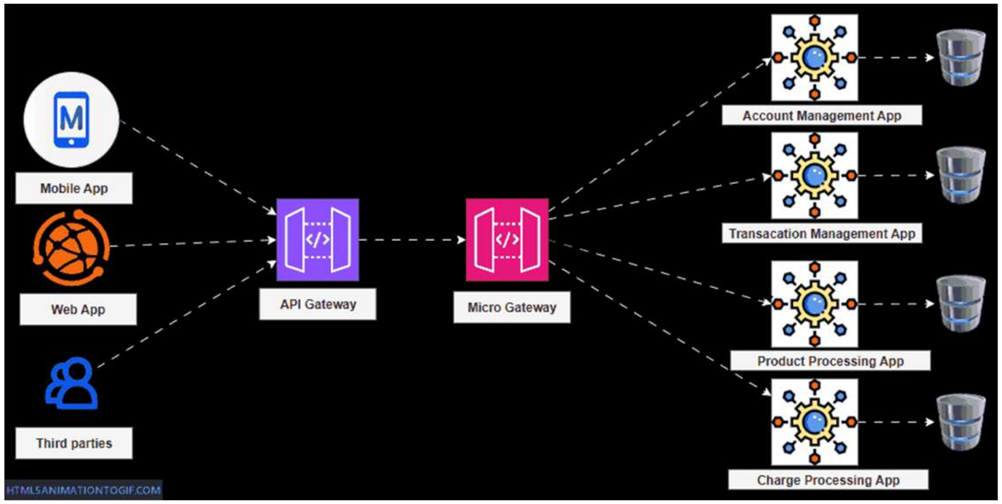
Enhanced Security Features

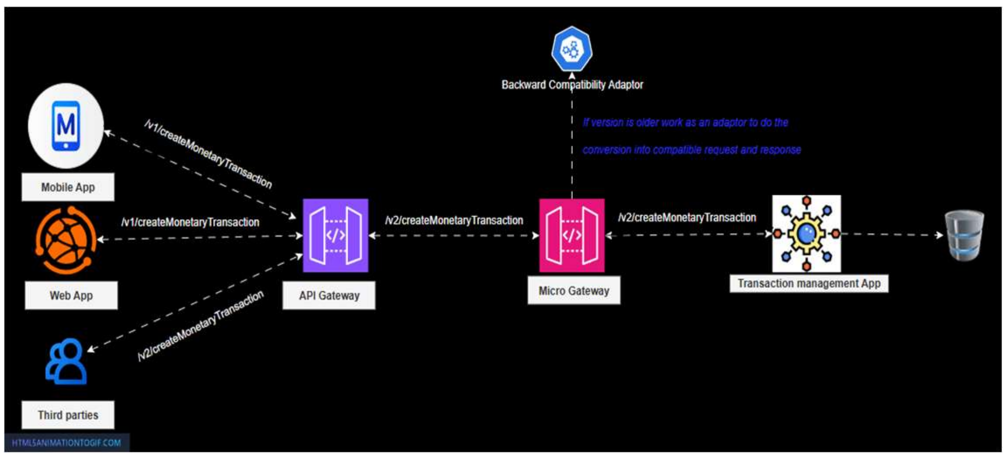
API Gateways offer security features like authentication, rate limiting, and detailed logging to protect applications.

Analytics and Optimization

Comprehensive analytics in API management help optimize performance, enhance user experience, and maintain compliance with security standards.

Real Time Integration - API Gateway Architecture





Hybrid gateway strategy combining centralized control with distributed micr
gateway flexibility for optimal performance and security.

Centralized API Gateway

Single entry point for external clients. Handles authentication, rate limiting, global routing, unified observability, and compliance enforcement.

Micro-Gateways

Domain-specific control for high-risk banking services. Provides latency optimization, fault isolation, service-level policies, and versioning support.

Real Time Integration - API Backward Compatibility

Supporting Legacy / Current Integration

Backward compatibility ensures older client applications keep functioning with new API versions, preventing disruption for end users.

Incremental Changes and Maintenance

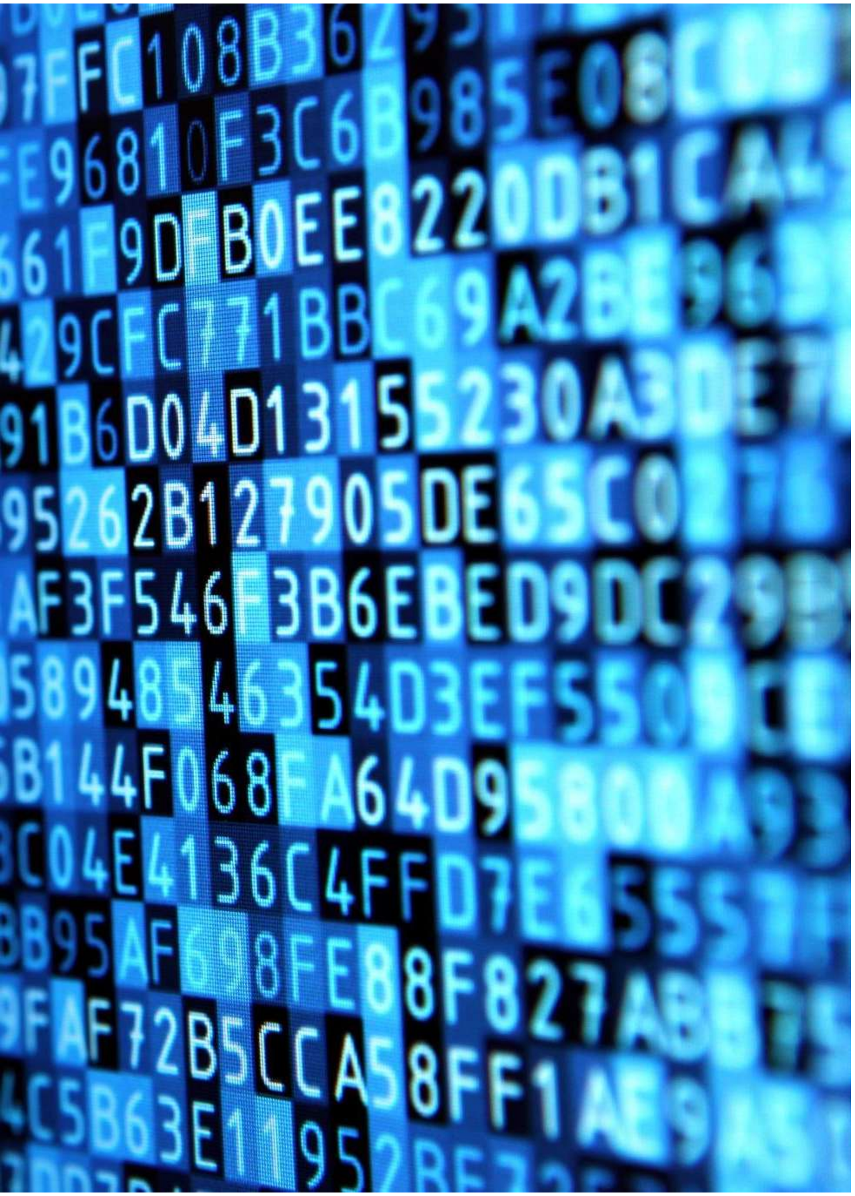
Maintaining existing endpoints and data formats allows changes to be introduced gradually, reducing risk of breaking integrations.

Versioning and Deprecation Policies

Careful API versioning and clear deprecation policies help manage updates efficiently and communicate changes to users.

Thorough Testing for Reliability

Comprehensive testing is essential to ensure new API updates do not disrupt service for existing users.



Real Time Integration - API Semantic Versioning

Semantic Versioning Structure

Semantic versioning uses MAJOR.MINOR.PATCH format to identify API release changes and updates clearly.

Major Version Changes

Incrementing the MAJOR version signals breaking changes which may affect backward compatibility in API integrations.

Minor and Patch Updates

MINOR versions add new features without breaking existing integrations, while PATCH fixes bugs ensuring smoother API performance.

Real Time Integration - API Semantic Versioning

AP Versioning strategy ensures smooth API evolution while maintaining client compatibility.

1

MAJOR Version

Breaking changes requiring client updates. URL-based versioning: `/v1/customers` → `/v2/customers`

2

MINOR Version

Backward-compatible enhancements. Header-based: `X-API-Version: 1.2.3`

3

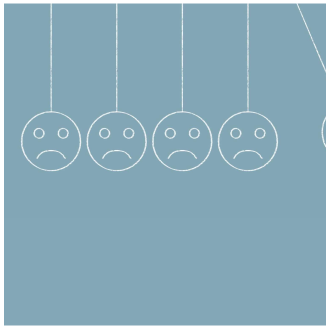
PATCH Version

Bug fixes and performance improvements with no interface changes

Real Time Integration - API Exception Handling

Capturing Error Responses

Detect and log errors such as HTTP status codes and error messages to monitor integration issues efficiently.



Consistent Exception Strategies

Implement uniform handling of exceptions to improve user experience and maintain predictable system behavior.

Facilitating Debugging

Helps developers quickly identify and resolve integration problems during API communications.

Financial Transaction Tracing and Reconciliation

Transaction Tracing Across Systems

Tracing payments across multiple systems ensures each transaction is recorded accurately and completely in financial records.

Automated Reconciliation Checks

Reconciliation matches records between systems, revealing and resolving discrepancies to maintain trustworthy financial data.

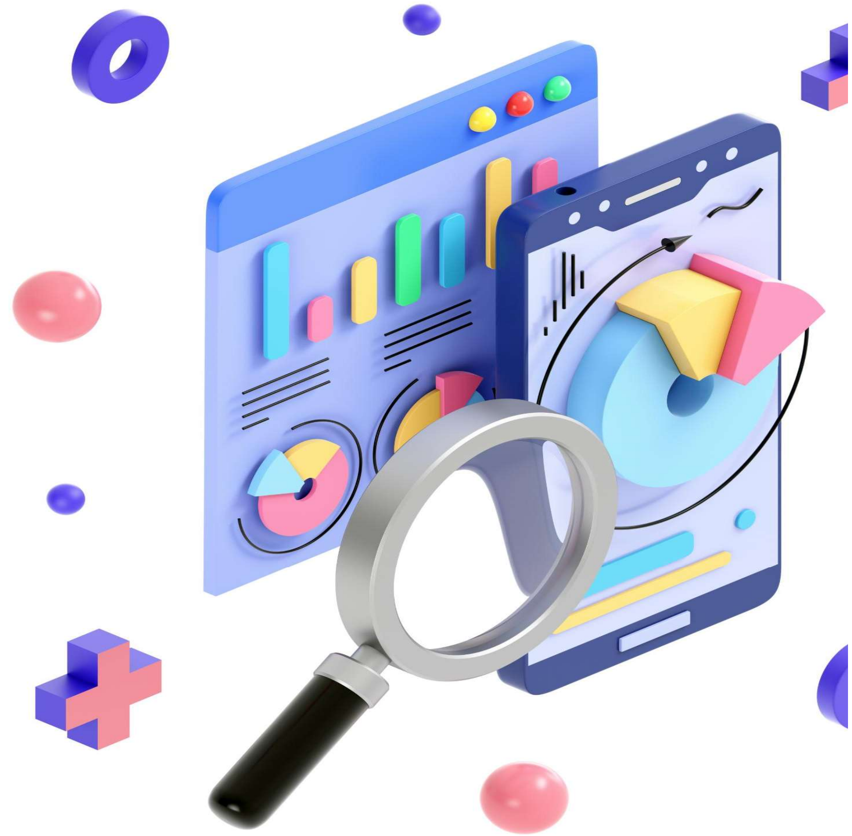
Transaction Reference ID's along with other attributes such as Account Number, Date, Amount … are used to Trace the

Automated Reconciliation Tools perform Reconciliation of large volumes of data and Report Exceptions (through Alerts / Operational Reports) if any mismatches are found

Reconciliation Exception Management

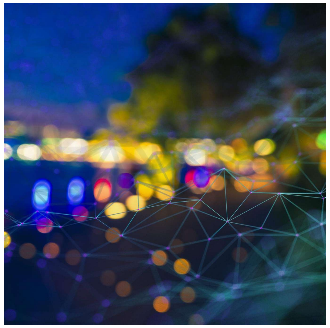
Exceptions are categorized, assigned priority levels and routed for investigation and resolution.

Note: An End to End Reconciliation Management Process needs to included as a part of the Design of Integrating between Systems. The Integration Pattern can be Real Time, Near Realtime or Batch/Bulk.



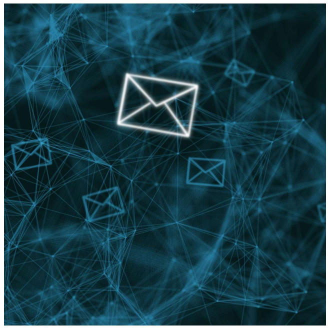
Near Real Time Integration - Events

Near Realtime Integration - Events



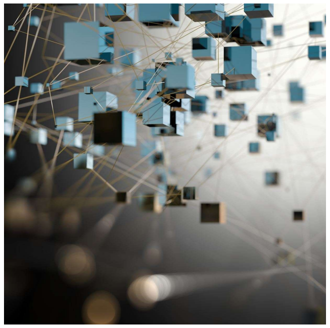
Event-Driven Architecture

Utilizes event-driven design to trigger instant system communication when changes occur, leading to rapid integration.



Event Management Platforms (Event Brokers)

Event Management platforms capture and transmit events, enabling near Realtime data synchronization between applications.



Scalable and Efficient Data Flows

This approach supports efficient data flows and seamless scalability across diverse and growing application environments.

Near Realtime Integration - Event Brokers

Facilitating Communication

Event brokers manage and transmit events between applications, ensuring seamless and efficient communication across systems.

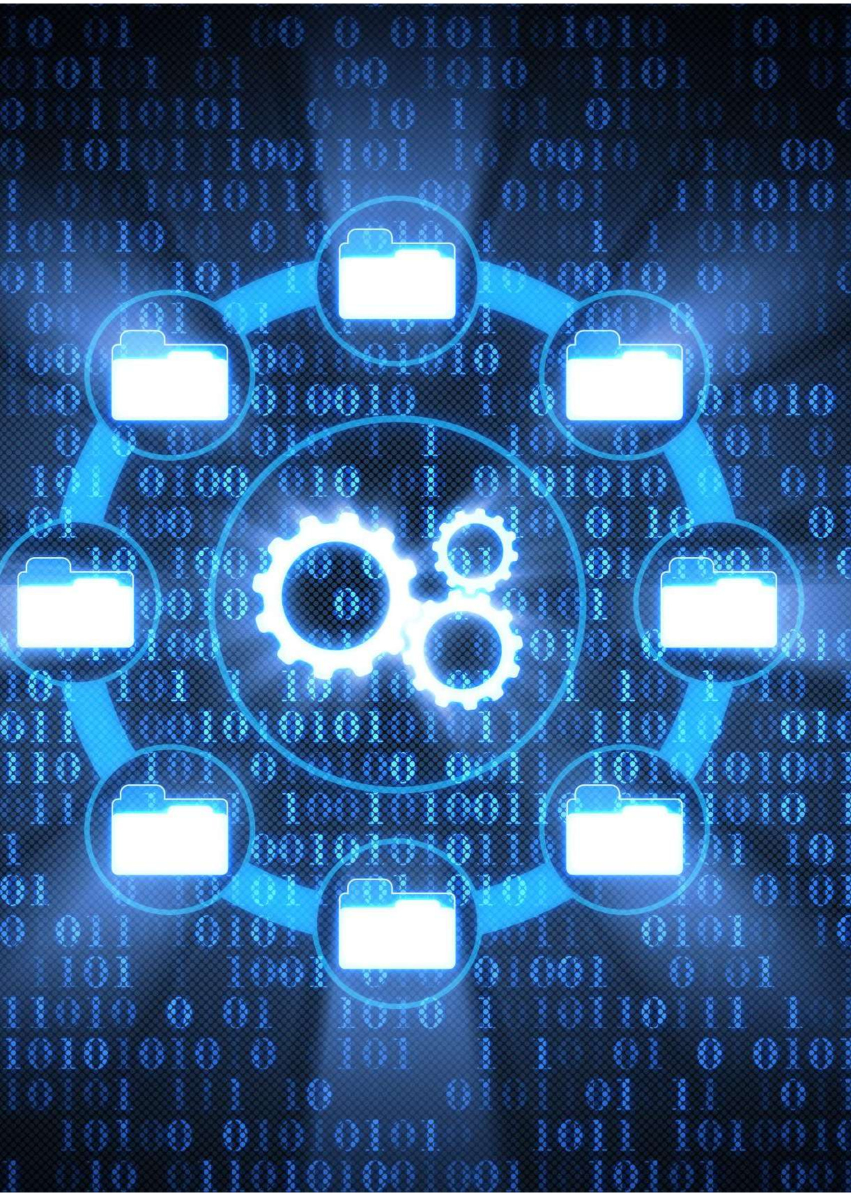
Decoupling Components

They decouple event producers from consumers, allowing independent scaling and greater system flexibility.

Supporting Real-Time Data

Event brokers enable real-time data flows, vital for modern microservices architectures and integrated digital solutions.

Note: Kafka is the leading Industry Event Broker



Near Real Time Integration – Event Backward Compatibility

Importance of Backward Compatibility

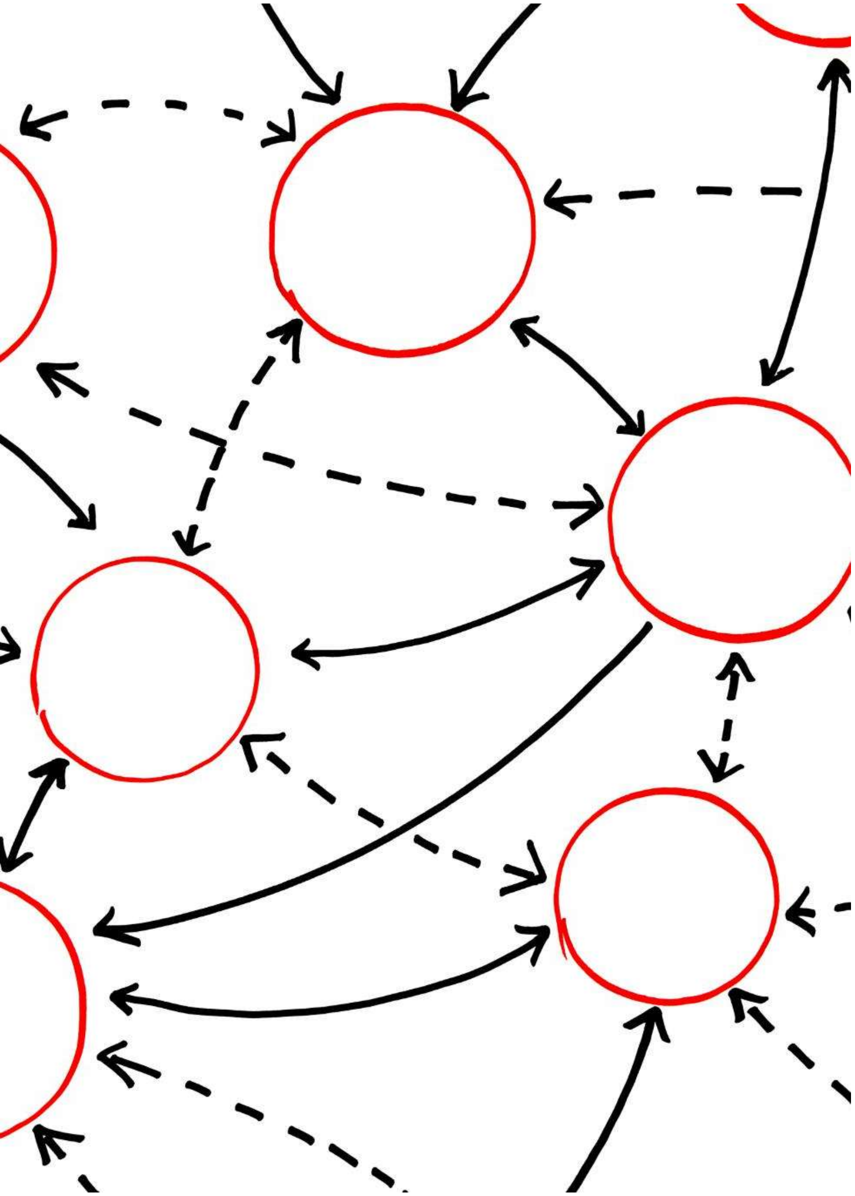
Backward compatibility ensures older event formats are processed reliably, helping systems avoid failures and maintain stability.

Key Compatibility Strategies

Versioning event schemas, supporting deprecated fields, and thorough testing protect legacy integrations as event structures evolve.

Reducing Downtime and Errors

Maintaining event compatibility during system updates minimizes downtime and prevents errors, ensuring smooth operation for users.



Near Real Time Integration – Event Semantic Versioning

Major Changes Indicate Breaks

Major version changes signal breaking changes in event schema or behavior, requiring consumers to adapt integration.

Minor Changes Add Features

Minor changes are backward-compatible enhancements, allowing new functionality without disrupting existing consumers.

Patch Changes Improve Stability

Patch changes fix minor issues, improving stability and reliability without affecting compatibility for event consumers.

Near Real Time Integration - Event Exception Handling

Robust Error Logging

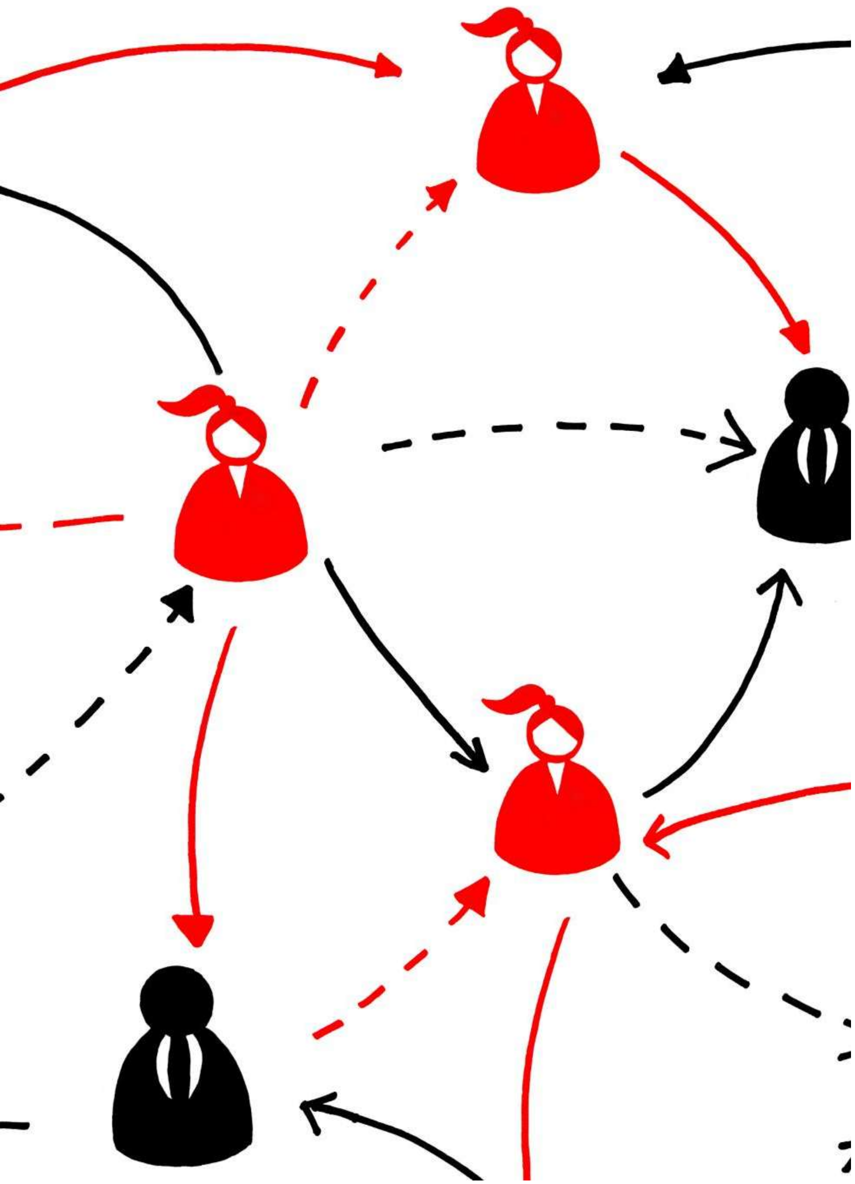
Detailed error logging helps capture, diagnose, and resolve failures that occur during event consumption and processing.

Retry and Dead-Letter Strategies

Using retry mechanisms or dead-letter queues ensures unprocessable events are managed without system disruption.

Meaningful Notifications and Alerts

Timely notifications and alerts enable operations teams to respond quickly and resolve exceptions effectively.



Near Real Time Integration - Event Reconciliation between Producers and Consumers

Ensuring Data Consistency

Reconciliation checks data between event producers and consumers to ensure information remains consistent across systems.

Detecting Event Issues

The process identifies lost, duplicated, or unprocessed events, helping maintain data integrity and minimize errors.

Supporting Reliable Integration

Effective reconciliation supports reliable integration between systems, building trust and stability in data-driven processes.

Batch/Bulk Integration - Files

Batch/Bulk Integration - Files

Structured File Formats

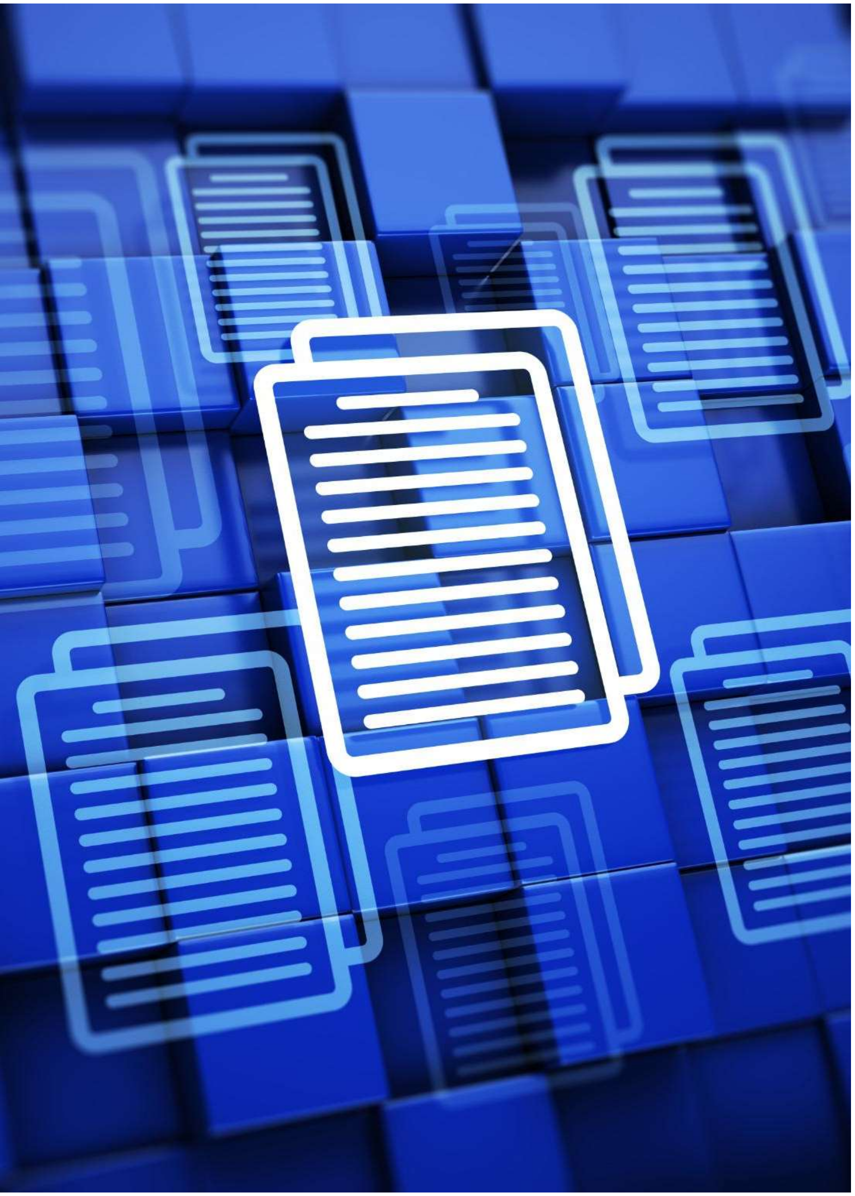
CSV, XML, and JSON are common formats for file-based data interfaces, supporting reliable data exchange between systems.

Batch Processing Integration

File data interfaces are ideal for batch processing, enabling the integration of data across diverse platforms efficiently.

Security and Integrity

Standardization, error handling, and security measures are essential to maintain the accuracy and protection of file-based data transfers.



Batch/Bulk Integration - File Headers and Footers

Role of File Headers

Headers store essential metadata at a file's beginning, including type, size, and format, guiding software interpretation.

Purpose of File Footers

Footers are found at the end of files and often include closing data or checksums for validation and integrity checks.

Ensuring Data Integrity

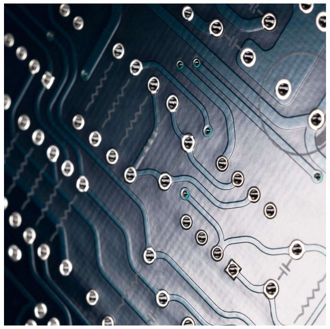
Headers and footers together enable efficient file processing and help maintain data integrity across different applications.

Industry Standard Interfaces

ISO8583

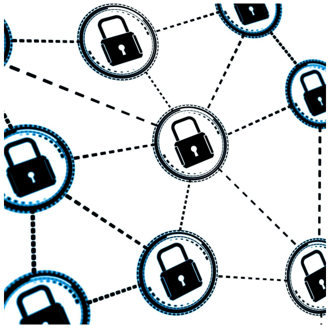
Standardized Card Transaction Messaging

ISO8583 standardizes how financial card transaction messages are structured and exchanged between payment systems.



Structured Data Elements

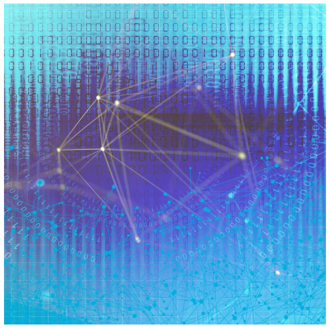
The standard uses data elements for key information, such as transaction amounts, card numbers, and processing codes.



Ensuring Interoperability

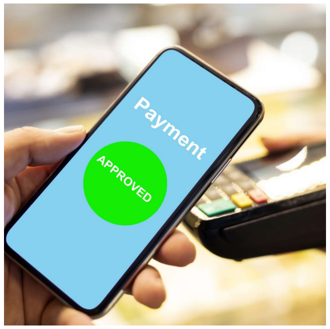
ISO8583 ensures seamless interoperability between different banks and payment networks across the globe.

ISO 20022



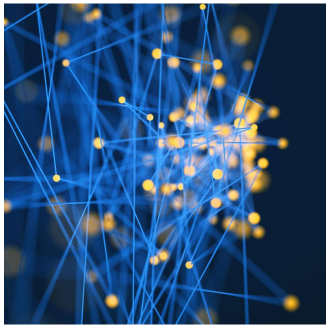
Structured Financial Messaging

ISO 20022 enables rich, structured data in financial messaging, making transaction information clearer and more useful.



Improved Payments Efficiency

By standardizing communication, ISO 20022 enhances payment transparency and operational efficiency for institutions worldwide.



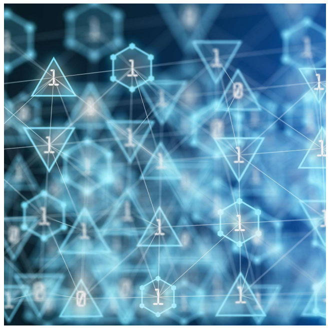
Enhanced Interoperability and Compliance

Global adoption supports interoperability, reduces errors, and strengthens regulatory compliance across the financial sector.

BIAN Framework

Standardized IT Architecture

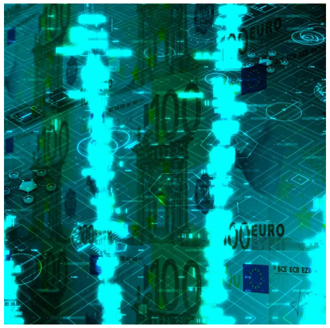
BIAN (Banking Industry Architecture Network) introduces a standardized IT framework for banks, promoting consistent technology practices and improved flexibility for change.



Service Domain Approach

BIAN defines specific service domains for key banking functions, supporting easier and more effective system integration.

The modular design of BIAN APIs allows banks to integrate new technologies easily and scale services as needed



Benefits for Financial Institutions

Adopting BIAN helps banks to become Vendor Agnostic and also accelerate their API Enablement and Adoption

Open Banking Overview

Secure Data Sharing

Open Banking allows banks and third-party providers to share financial data securely, enabling new financial services for customers.

Innovation and Customer Choice

Customers benefit from increased competition and access to innovative services.

APIs empower developers to build new financial products and services, encouraging creativity and competition in the industry.

Regulatory Protection

Regulatory frameworks and user consent protocols ensure customer protection, data privacy.

Open Banking Worldwide

India:

UPI

Unified Payments Interface (UPI) enables instant and easy digital transactions, transforming India's payment ecosystem.

Consolidated Financial Information

Account Aggregator APIs help customers access and manage consolidated financial details from multiple sources in one place.

Automated Bill Payments

Bharat BillPay APIs facilitate automatic bill payments, improving convenience and efficiency for users

UK

UK regulations require banks to share data securely with licensed providers, encouraging financial innovation and new services.

Australia

Australia empowers consumers to access and control their financial data, giving them greater choice and transparency.

Brazil

Brazil is implementing Open Banking in phases to boost competition and enhance customer services in the financial sector.